

All thresholds use PERCENTILE RANK (3yr rolling, not z-score). DUAL paradigm = absolute floor + percentile rank. ROC = rate-of-change %. ABS = absolute level. RATIO = ratio threshold.

#	Variable	ID	Source	Paradigm	RARE threshold	EXTREME threshold	Combos	Inflation adj?
1	GS Financial Conditions (NFCI)	FCI	FRED: NFCI	DUAL	+0.3 SD ≈ 80th pctile 1973–2026	+ 0.8 SD ≈ 95th pctile	A, E	No — index already normalised
2	HY Credit Spreads OAS	HY	FRED: BAMLH0A0HYM2 [SSI uses HYG/LQD — don't double-count]	DUAL	>400bps OR >+1.0 SD 3yr	> 500bps OR >+2.0 SD	A, B, F, G	No — spreads are relative
3	Fed Balance Sheet MoM Δ%	WALCL	FRED: WALCL	ROC	±0.8% MoM	±2.0% MoM	A, C	No — % change is scale-invariant
4	USD/CNH Exchange Rate	CNH	Yahoo: USDCNH=X	ROC	±1.5% in 4 weeks	±3.5% in 4 weeks	A, C, G	No — % change is scale-invariant
5	WTI Oil 4-week % change	WTI	Yahoo: CL=F	ROC	±6% in 4 weeks	±10% in 4 weeks	C	No for % change. Yes for absolute level comparisons across decades — use WTI/CPI deflated.
6	VIX Level	VIX	CBOE: ^VIX [Runic only — excluded from SSI Layer 2]	DUAL	>25 AND >80th pctile 1990–2026	> 35 AND >95th pctile	B, D, G	No — vol is vol
7	VIX Term Structure VIX3M÷VIX	VXTS	Yahoo: ^VIX3M / ^VIX [Runic only — excluded from SSI Layer 2]	RATIO	<0.95 (backwardation) or >1.10 (deep contango)	<0.85 or >1.20	D, G	No — ratio is dimensionless
8	CFTC Net Spec Positioning	COT	CFTC.gov TFF report Lev_Money + Asset_Mgr cols	PCTILE	<15th or >85th pctile 3yr rolling	<5th or >95th pctile	B, D, E, F	No — percentile already normalises
9	10Y–2Y Yield Curve	CURVE	FRED: T10Y2Y	DUAL	<–30bps OR steepening >15bps/4wk post-inversion	<–80bps OR steepening >40bps/4wk	A, E	Use REAL 10Y for structural level analysis. Nominal for rate-of-change signals.
10	CPI Surprise vs Consensus	CPI	BLS vs Investing.com economic calendar	ABS	±0.2pp single print	±0.4pp OR 2× consecutive hot	C	Already measures surprise — inherently real
11	Gold/Silver Ratio (GSR)	GSR	FRED: GOLD÷SILVER or Yahoo GC=F÷SI=F	ROC	4wk Δ ≥±5% [nominal ratio — no inflation adj]	4wk Δ ≥±8%	A+	No — ratio normalises itself. GSR is already a relative price.
12	CAPE / Shiller P/E	CAPE	multpl.com or FRED	ABS	>28x or <16x ≈85th/10th pctile 1881–now	>32x or <12x ≈95th/5th pctile	E	Already inflation-adjusted by construction

NAMED COMBO REGIMES — 7 pre-validated combinations. All others (291) are auto-discovered by the engine at ≥80% hit rate gate.

COMBO	Name	Variables required (ALL at RARE+ simultaneously)	Direction	Horizon	Hit rate	Avg return	Key threshold notes	Key historical instances (validated)
A	Global Liquidity / FCI Regime	FCI + HY direction (both tightening=FEAR; both easing=BRAVE) + WALCL + CNH (≥2 of 4 required)	BRAVE or FEARFUL	3–6m	~78%	BRAVE +12% FEARFUL –8%	FCI is the master switch. HY confirms. WALCL confirms QE/QT. CNH adds EM stress.	GFC QE (2009), COVID QE (Mar 2020), Fed pivot (Jan 2019), Hike onset (Mar 2022)
B	Maximum Capitulation / Blood in Streets	VIX >25 + 80th pctile AND HY >400bps AND CFTC <15th pctile (ALL 3 simultaneously — doubles/singles are WATCH only)	BULLISH (contrarian)	3m	87–91% (91% cutting cycle 68% hiking cycle)	+14% avg 3m	■ VIX REGIME MULTIPLIER BYPASSED when Combo B active. Never reduce size on confirmed B.	Aug 2015, Feb 2016, Dec 2018, Mar 2020 (VIX 82), Oct 2022 ✓ VALIDATED: Oct 13, 2022 — SPX +19% 3m later
C	Stagflation / Energy Shock	WTI >+10% in 4wks AND CPI ≥+0.2pp surprise AND WALCL flat/shrinking Duration: SHORT <6wks, MEDIUM 6–16wks, LONG >16wks	BEARISH	1–6m	~83%	Neg 1–6m	CANCEL only when BOTH WTI 4wk Δ <+5% for 4 consec weeks AND CPI prints not hot. Cancel requires BOTH legs to clear.	Jun 2008 (\$147), Jun 2022 (CPI 9.1%), Apr 2025 (tariff), Apr–May 2026 Iran war (MEDIUM duration wk 11 as of May 26, 2026)

D	FOMO Top / Euphoria Tactical	VXTS ratio >1.10 (deep contango) AND CFTC >85th pctile AND VIX <18 (short-horizon only — needs Combo E to confirm structural top)	BEARISH (tactical)	3–10 days to 4wks	~72–85%	–3 to –5% / 5 days	VXTS >1.10 = near-term vol extremely compressed (complacency). Long-run avg = 1.07–1.10. Today ~1.25 = Combo D partial.	Jan 2018 (Volmageddon), Feb 2020 pre-COVID, Nov 2021, Apr 2026 BTIG Today VXTS ~1.25 = at WATCH level
E	Valuation Extreme / Slow Burn Top	CAPE >28x AND FCI easy (NFCI <–0.3) AND CFTC >80th pctile (2 of 3 at RARE+ required)	BEARISH (structural)	6–18m	~73%	Neg fwd 12m	Long-horizon warning — NOT a sell signal alone. Acts as MULTIPLIER on Combo D. Markets stay expensive for years. CAPE >32 = EXTREME.	Mar 2000 (CAPE 44x), Oct 2007 (27x) Nov 2021 (38x) Today CAPE 42.0x → EXTREME tier
F	Recovery / Reentry Signal	SPX reclaims 50-Week MA with ≥+3% weekly gain WHILE CFTC ≤50th pctile (not yet extreme long — fuel for forced covering)	BULLISH	3–6m (26-week window)	~78% (91% in QE regime)	+9.46% avg 6m (16 instances 1932–2026)	Prior drawdown below 50WMA required (no minimum duration specified). CFTC condition critical: hit rate drops ~20pp if CFTC already extreme long at reclaim. ✓ VALIDATED: Jun 29, 2020 — SPX +18.3% 6m later	Dec 1932, Jun 1950, Jun 2020 (+18.3%) Oct 2023 (+17%), Mar 30, 2026 (active wk 8 of 26 as of May 26, 2026)
G	Hidden Stress / Credit-Vol Divergence	VXTS <1.0 (backwardation — near-term fear > medium-term) AND HY widening >+30bps in 4wks WHILE VIX <20 (stress in credit not yet priced in vol)	WARNING (leading)	Leads vol spike 3–6 weeks	~75% lead	Not a return predictor — a timing warning	Most powerful early warning. Fires BEFORE the event. Watch for G→B cascade. Apr 7, 2025: VIX ~52, VIX/VIX3M in backwardation — G fired ~3wks prior.	pre-Aug 2015, pre-Dec 2018 pre-COVID (Feb 2020) Apr 2025 episode (now resolved)

SSI CONFIDENCE MULTIPLIER	MACRO REGIME ANNOTATION (auto-classified via Claude API)	CURRENT STATUS — MAY 26, 2026	METHODOLOGY NOTE — THRESHOLD JUSTIFICATION
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Scales every combo score after it fires. • SSI CONFIRMED (≥2 of 4 agree): ×1.20 • SSI PARTIAL (1 of 4): ×1.00 • SSI UNCONFIRMED (0 of 4): ×0.80 4 real-time SSI signals: VIX/VIX3M ratio HYG/LQD ratio DBMF 21-day beta vs SPY CNN F&G; NOT standalone buy/sell. Modifies an already-fired combo only. UNCONFIRMED ≠ false alarm. It means: 'Macro signal present, sentiment not yet responding — reduce size, do not ignore.' Every combo fire tagged with 5 dimensions (generated by Claude API — NOT manual): 1. fed_cycle: HIKING_EARLY / HIKING_LATE CUTTING_EARLY / CUTTING_LATE / PAUSING QE / QT 2. curve_regime: INVERTED / FLAT / STEEPENING / NORMAL 3. geo_overlay: NEUTRAL / TRADE_WAR / SANCTIONS / REGIONAL_WAR / PANDEMIC / FINANCIAL_CRISIS 4. val_regime: EXTREME / ELEVATED / FAIR / CHEAP 5. liquidity: GLOBAL_EASY / GLOBAL_TIGHT Hit rate SQL (regime-adjusted): SELECT AVG(CASE WHEN spx_3m>0 THEN 1.0 ELSE 0 END) FROM combo_fires cf JOIN forward_returns fr ON cf.id=fr.combo_id WHERE cf.runic_combo='B' AND json_extract(cf.macro_regime, '\$.fed_cycle') LIKE 'CUT%' NFCI: -0.523 → LOOSE. Not at RARE (+0.3). HY OAS: ~305bps → NORMAL (~20th pctlile). CAPE: 42.04x → EXTREME (>32 threshold). VIX: 16.7 → NORMAL. Below RARE (25). VXTS ratio: ~1.25 → WATCH (D thresh 1.10). 10Y-2Y curve: +43bps → STEEPENING. WTI 4wk Δ: ~-17% (↓ from \$117 peak, now \$91 — C CANCELLATION APPROACHING) CFTC: needs Friday pull (est. elevated long). GSR: needs Friday pull. DATE CORRECTIONS: Apr 7, 2025 = tariff shock VIX 52 low (Combo B+G — now resolved) Mar 30, 2026 = Combo F fire (50WMA reclaim) Apr-May 2026 = Iran war WTI spike (Combo C) May 26, 2026 = TODAY ACTIVE: C (wk 11, MEDIUM) + E (partial) + F (wk 8 of 26) WATCH: D (VXTS 1.25 approaching 1.10) DOMINANT: TACTICAL FEARFUL / STRATEGIC BRAVE METHODOLOGY: With only 5-12 extreme instances per combo since 1985, formal optimisation would overfit. Correct approach: set thresholds at meaningful statistical levels (80th-95th pctlile), verify they classify ALL known historical extreme episodes correctly, then measure forward returns from those episodes. 8 confirmed Combo B instances since 1990 → 7/8 positive 3m later = 87.5% hit rate. Small N, 100% consistent with theory. That IS the validation. PERCENTILE not z-score: Financial distributions are fat-tailed (leptokurtic). VIX, HY, CFTC all have fatter tails than normal. Z-score understates extremes. Percentile rank is distribution-agnostic. Use for all 12 variables. OVERLAP RULE: HY OAS (FRED) feeds Runic. HYG/LQD ratio feeds SSI. SAME underlying — do NOT double-count in combined score. VXTS allocated to Runic only (Combo D/G). Excluded from SSI Layer 2.			
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